Ananth Jonnavittula

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Summary

Robotics researcher with 10+ years across industry and academic research, building robot foundation models and agentic AI systems that run on real hardware. Recent work spans VLA policy training and deployment, LLM agent-in-the-loop task planning for long-horizon humanoid tasks, imitation and sim-to-real reinforcement learning, and perception-aware whole-body control. Track record of taking learned policies from simulation and demonstration data to real humanoids and production factory floors, with first-author publications at ICRA, IROS, and ACM THRI.

Technical Skills

Foundation Models & Agents: VLAs, LLM Agents, VLMs, SAM2/SAM3, DINOv2
Robot Learning: Imitation Learning / DAgger, Sim-to-Real RL, Whole-Body Control, Loco-Manipulation
Simulation & ML: Isaac Lab, Isaac Sim, PyTorch, TorchScript, Blender / BlenderProc
Robotics & Perception: ROS 2, Nav2, VR Teleoperation, OpenCV, 3D Scene Graphs
Languages: Python, C++
Platforms: Unitree G1 and custom humanoids, Universal Robots, ABB, FANUC, KUKA, Franka Emika, Fetch

Education

Virginia Tech **Aug 2020 – Aug 2024**
PhD in Mechanical Engineering (Robot Learning) *Blacksburg, VA*

Worcester Polytechnic Institute **Aug 2015 – May 2017**
MS in Robotics Engineering *Worcester, MA*

SASTRA University **Jun 2011 – May 2015**
B.Tech in Electronics and Instrumentation *Thanjavur, India*

Work Experience

Senior Research Engineer **Sep 2025 – Present**
Samsung Research America *Mountain View, CA*

- Developed a perception-aware whole-body loco-manipulation policy for humanoid object carrying through tight spaces
- Built a perception-aware whole-body policy that resets a humanoid to its initial state in confined, cluttered spaces
- Co-developed C-SOP, an LLM agent-in-the-loop architecture that compiles plain-language procedures into a state machine and orchestrates perception, navigation, and VLA manipulation on a Unitree G1 humanoid
- Led agent-to-robot integration and whole-body control for C-SOP: tool interface dispatching LLM function calls over ROS 2 to scene-graph, Nav2 navigation, and manipulation skills on the humanoid
- Trained and deployed GR00T N1.7-based VLA policies on humanoid hardware using SONIC for whole-body control
- Engineered Isaac Lab simulation assets, environments, and training and evaluation pipelines for humanoid whole-body control policies, benchmarking policies in simulation before hardware deployment

AI Researcher **Dec 2024 – Sep 2025**
Foundation Future Industries Inc *San Francisco, CA*

- Trained RL locomotion controllers in Isaac Sim for a custom humanoid and transferred them sim-to-real onto hardware
- Deployed Diffusion Policy imitation learning for dexterous manipulation in production factories, improved via DAgger
- Built the Isaac Sim simulation environment and robot model for the company's custom humanoid
- Created DINOv2 + SAM2 segmentation and pose estimation for uncommon automotive parts

Graduate Research Assistant **Aug 2020 – Aug 2024**
Virginia Tech *Blacksburg, VA*

- Developed VIEW, a one-shot visual imitation learning method that learns long-horizon robot tasks from a single third-person video using pretrained vision models and trajectory waypoints
- Implemented Soft Actor-Critic reinforcement learning for manipulation of YCB objects
- Engineered an inverse reinforcement learning algorithm to learn rewards from imperfect user demonstrations
- Developed a variational autoencoder (VAE)-based algorithm for imitation learning
- Conducted stability analysis of imitation learning methods for shared autonomy (IROS Best Student Paper Finalist)

Research and Development Intern **Jun 2022 – Aug 2022**
ABB Inc *San Jose, CA*

- Implemented and benchmarked stereo-vision and RGB-D methods for instance segmentation
- Architected a detectron2 instance segmentation model for small-parcel singulation
- Built a BlenderProc pipeline to generate synthetic datasets for model training
- Exported trained models from Python to C++ with TorchScript for deployment on industrial robots

Robotics/Vision Engineer **Jun 2017 – May 2020**
Parker Hannifin Corp. *Haverhill, MA*

- Engineered an automated palletizing cell handling 100+ SKUs to reduce labor costs
- Designed and implemented an automated laser marker that doubled throughput across multiple manufacturing cells
- Developed a UR5 robotic cell with cameras for part recognition and filter end capping
- Engineered and deployed a urethane end-capping cell using FANUC robots

Engineering Intern **Jan 2017 – Apr 2017**
Parker Hannifin Corp. *Haverhill, MA*

- Conducted feasibility analysis for automated end capping using collaborative robots
- Programmed controllers for automated part feeding using vibratory feeders
- Implemented image recognition with Keyence CV-X vision systems to detect part orientation

Graduate Research Assistant **Jan 2016 – May 2016**
Worcester Polytechnic Institute *Worcester, MA*

- Analyzed range-of-motion and kinematic requirements for a 2-DOF hydro-muscle actuated leg
- Designed a coupling mechanism that locks the leg in pose during a serious failure (led to US patent)
- Developed closed-loop control for leg actuation using four pairs of hydro-muscles
- Built Arduino-based controllers for the hydraulics and coupling mechanism

Publications

CarryEverywhere: Towards Humanoid Payload Transport across Cluttered and Dynamic Environments **2026**
In Submission
*Gershom Seneviratne, **Ananth Jonnavittula**, Bharath Masetty, Swathi Jadav, and Kris Hauser*

C-SOP: Continuous Standard Operating Procedure-Guided In-Home Agentic Robot **2026**
Under Review
*Xiao Liu, Holly Dinkel, **Ananth Jonnavittula**, M. Oller Oliveras, D. Patel, S. Jadav, Y. Shirai, S. Habibian, and Kris Hauser*

VIEW: Visual Imitation Learning with Waypoints **2025**
Autonomous Robots (Springer)
***Ananth Jonnavittula**, Sagar Parekh, and Dylan P. Losey*

SARI: Shared Autonomy across Repeated Interaction **2024**
ACM Transactions on Human-Robot Interaction
***Ananth Jonnavittula**, Shaunak A. Mehta, and Dylan P. Losey*

Communicating Robot Conventions through Shared Autonomy **2022**
IEEE International Conference on Robotics and Automation (ICRA)
***Ananth Jonnavittula** and Dylan P. Losey*

Here's What I've Learned: Asking Questions that Reveal Reward Learning **2022**
ACM Transactions on Human-Robot Interaction
*Soheil Habibian, **Ananth Jonnavittula**, and Dylan P. Losey*

Learning to Share Autonomy Across Repeated Interaction **2021**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) *Best Student Paper Finalist*
***Ananth Jonnavittula** and Dylan P. Losey*

I Know What You Meant: Learning Human Objectives by (Under)estimating Their Choice Set **2021**
IEEE International Conference on Robotics and Automation (ICRA)
***Ananth Jonnavittula** and Dylan P. Losey*

Patents

Biologically inspired joints and systems and methods of use thereof **2020**
US 10,632,626 B2